Assignment 2

COL761 2026

Instructions

📅 Deadline: **March 16, 2026.**

🚫 Anti-Plagiarism Policy: Plag detection equals F-grade in the course.

✍️ Cite your sources in the report.

📊 Datasets: For Q1, use API (see below). For Q2, links provided within the question.

🐍 `Python 3.10` and `C/C++` are allowed.

Submission related instructions:

💻 Add to GitHub.

1️⃣ Only one person per team should make the submission.

➕ More in the document.

Q1 k-means clustering in high-dimensions [20 marks]

Let $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$ with $d > 3$ denote a given dataset. Consider the k-means objective

$$\min_{\{\mathcal{C}_j\}_{j=1}^k} \sum_{j=1}^{k} \sum_{x_i \in \mathcal{C}_j} \|x_i - \mu_j\|^2,$$

where μ_j is the empirical mean of cluster $\mathcal{C}_j$. Assume Euclidean distance.

Computational Task.

For each $k \in \{1, 2, \ldots, 15\}$:

(a) Compute an optimal solution of the k-means objective.
(b) Record the corresponding objective value.

Plot the objective value as a function of k. Based on this empirical evidence, determine a suitable choice of k.

Written Report (maximum 1.5 pages, excluding references).

Your report must address the following:

(a) The reasoning that leads to your selected value of k.
(b) The structural and statistical assumptions underlying k-means.
(c) Limitations of the above model selection procedure.
(d) Possible methodological improvements or alternative validation principles.
(e) A discussion of conditions under which k-means is not an appropriate model for the data. Illustrate your argument using well-known failure modes of k-means and provide a principled procedure for detecting such cases in practice.

Figures should be clearly labeled and incorporated into the report.

Clarity, precision of reasoning, and depth of analysis will be weighted more heavily than implementation detail.

Submission Requirements

Code

Submit a script `Q1.py` as a front-end for our auto-eval to interact with. It should run in two modes, as follows

```
python3 Q1.py <dataset_num>
python3 Q1.py <path_to_dataset>.npy
```

In the first mode, it should load the dataset from the dataset API shared below, taking `dataset_num` as an argument (see listing 1 for details). In the second, it should read a `.npy` file. The `.npy` file supplied as input is guaranteed to contain an `np.ndarray` identical in structure, shape, and data type to the array assigned to `data` (i.e., the result of `np.array(data["X"])`) in line 16 of listing 1 below.

`Q1.py` should provide two outputs:

(a) A file named `plot.png` in the current directory, that contains the plot, and
(b) A single number in the `stdout` that corresponds to the optimal value of k.

Report

- Maximum length: **1.5 pages** (excluding references). Filename: `report_q1.pdf`.
- Figures must be readable and properly labeled.
- Emphasis will be placed on clarity of reasoning and critical analysis.

Directory Structure

```
A2
└── q1
    ├── Q1.py
    ├── plot.png # generated from Q1.py
    └── report_q1.pdf
```

Dataset details You will be working with dataset specific to your kerberos id, which can be accessed as shown below.

Note: The following is only accessible from IIT network.

Listing 1: How to get the dataset

```python
1  import urllib.request
2  import json
3  import numpy as np
4
5  def main():
6      url = "http://10.208.23.248:3000/dataset?student_id=cs5210602&dataset_num=1"
7      # url = "http://10.208.23.248:3000/dataset?student_id=cs5210602&dataset_num=2"
8      # dataset_num can take values either 1 or 2
9
10     with urllib.request.urlopen(url) as response:
11         # Read the response and decode the bytes to a string
12         raw_data = response.read().decode('utf-8')
13         # Parse the JSON string into a Python object
14         data = json.loads(raw_data)
15
16     data = np.array(data["X"])
17     print(data)
18
19 if __name__=="__main__":
20     main()
```

Note: This assignment evaluates both implementation and your ability to reason about clustering behavior in high-dimensional spaces.

Q2 Route Blocking for Reducing Forest Fire Spread

Wildfires often spread across forest regions through natural terrain corridors such as dry vegetation paths, wind-aligned clearings, and forest access routes. Authorities can construct temporary firebreaks or block certain routes to slow down the spread. However, constructing firebreaks is costly, and only a limited number of routes can be blocked.

In this assignment, you will determine which forest routes should be blocked in order to minimize the expected spread of a wildfire.

Fire Spread Model

The forest is modeled as a directed graph $G = (V, E, p)$, where (a) $V = \{1, 2, \ldots, n\}$ represents forest regions (clusters of trees), (b) $E \subseteq V \times V$ represents possible fire-spread routes (directed source-destination pairs), and (c) $p : E \to (0, 1]$ assigns a propagation probability to each route. The set $\{u \mid (u, v) \in E\}$ is called the set of out-neighbors of u.

Let $A_0 \subseteq V$ denote the set of initially burning regions.

The fire evolves in discrete time steps:

- If region u is burning at time t, it gets exactly one opportunity to ignite each currently unburned out-neighbor v.
- The ignition attempt succeeds independently with probability $p_{(u,v)}$.
- If successful, v starts burning at time $t + 1$.
- Each tree cluster (node) burns at most once.
- The process stops when no new regions ignite.

Let A_∞ denote the set of burned regions at termination.

We are interested in the expected total number of burned regions when we block some routes:

$$\sigma(R) = \mathbb{E}[|A_\infty|],$$

where $R \subseteq E$ is the set of blocked routes and the fire spreads on the modified graph with blocked routes removed $G_R = (V, E \setminus R, p)$.

Route Blocking Problem.

Given a budget k, select a set $R \subseteq E$ such that

- $|R| = k$, and
- $\sigma(R)$ is minimized.

Since the process is stochastic, $\sigma(R)$ must be estimated using simulation.

Alternate Spread Model

Let h be a positive integer, then the h-hop spread model is defined by an additional constraint to the original fire spread model. Let

$$d(A_0, v) = \min_{u \in A_0} \text{graph-shortest-path-distance}(\text{src} = u, \text{dst} = v),$$

then no nodes outside the set $V_{h,A_0} = \{v \mid d(A_0, v) \leq h\}$ can be on fire at any time.

Your code will be evaluated on both spread models.

Tasks

Task 1: Optimization Formulation. [5 marks]

Formulate the route blocking problem as a combinatorial optimization problem. Clearly define the decision variables, objective function, and the constraints.

Task 2: NP-Hardness. [10 marks]

Prove that the route blocking problem is NP-hard. Your proof must be based on a polynomial-time reduction from a known NP-hard problem.

Task 3: Structural Properties. [10 marks]

Let $\sigma(R)$ denote the expected fire spread after blocking routes R. Analyze whether $\sigma(R)$ is:

1. Monotone [3],
2. Submodular [7],

Provide proofs or counterexamples.
Note: There are no marks for yes/no answers. Only the complete and correct proof will be assigned marks.

Task 4: Algorithm Design. [10 marks]

Propose an algorithm for selecting k routes to block.

- Give the pseudocode for your algorithm[5].
- Derive its computational complexity[2].
- Does the proposed algorithm provide any formal approximation guarantees, such as being provably within a bounded factor of the optimal solution? If so, please state the guarantee and provide proof. Note: There are no marks for yes/no answers. [3]

Task 5: Competitive Evaluation. [45 marks]

Implement your algorithm. Your program must:

- Estimate $\sigma(R)$ using randomized simulations,
- Scale to graphs with up to 15×10^4 regions,
- Respect the specified time limits.

The output must contain exactly k lines, each line containing a pair of integers representing a blocked route. `hops` can take values from this set: $\{-1\} \cup \{1, 2, 3, \dots\}$. When `hops` $= -1$, the standard spread model should be used, otherwise the alternate spread model with $h =$ `hops`. We will execute your solution using the following command:

Listing 2: Running command

```
bash forest_fire.sh <absolute_path_to_graph> <absolute_path_to_seed_set>
     <absolute_output_file_path> <k> <n_random_instances> <hops>
```

Estimation of Expected Spread.

To estimate $\sigma(R)$, generate r random realizations of the graph by independently sampling each remaining route according to its probability. For each realization, simulate the fire spread from A_0 and compute:

$$\frac{1}{r} \sum_{i=1}^{r} |A_\infty^{(i)}|.$$

Competitive Scoring

Task-5 carries 45 marks.

Let:

- $\sigma(\varnothing)$ be the expected number of burned regions when no routes are blocked,
- $\sigma(R)$ be the expected burned regions achieved by your submission,
- $\rho(R)$ denote the reduction ratio:

$$\rho(R) = \frac{\sigma(\varnothing) - \sigma(R)}{\sigma(\varnothing)}.$$

- ρ^* be the maximum reduction ratio achieved by any submission on that dataset.

Your score on a dataset will be:

$$45 \cdot \frac{\rho(R)}{\rho^*}.$$

Scores will be averaged across multiple public and hidden datasets.

For evaluation, $\sigma(\varnothing)$ will be computed once per dataset using a sufficiently large number of simulations and treated as fixed.

Two public datasets are provided below, with a few default paramters for you to get started. Please note that the parameters that we evaluate on may be different (so don't hardcode parameters).

- Dataset 1: $k = 50$, `n_random_instances` $= 50$, `hops` $= -1$ (fire spread is *unlimited*; regions at any distance from A_0 can be ignited.)
- Dataset 2: $k = 20$, `n_random_instances` $= 50$, `hops` $= 3$ (fire spread is *restricted to at most 3 steps* from the seed set A_0; regions more than 3 hops away cannot be ignited regardless of edge probabilities.)

Each dataset contains a graph file (V, E, p) and a seed set file A_0. The initial seed set A_0 may contain between 0 and 10 regions.

Evaluation scripts are provided in the folder `Eval`:

```
bash evaluate.sh <absolute_path_to_graph> <absolute_path_to_seed_set>
     <absolute_path_to_blocked_routes> <k> <n_random_instances> <hops>
```

The script computes $\sigma(\varnothing)$, $\sigma(R)$, and the corresponding reduction ratio using the same simulation protocol for all submissions. `hops` can take values from this set: $\{-1\} \cup \{1, 2, 3, \dots\}$. When `hops` $= -1$, the standard spread model should be used, otherwise the alternate spread model with $h =$ `hops`.

Note: Your submission must generate the `<absolute_path_to_blocked_routes>` file at the exact file path provided (not a folder path) in the running command (listing 2). The correct file format is expected so that it can be evaluated using the given script.

Submission Instructions

1. Submit a script `forest_fire.sh`.
2. Submit a `report_q2.pdf` containing (max. **5 pages**) containing solutions of tasks 1 through 4 and the evaluation metric achieved by your submitted code on the public datasets for task 5.
3. Use the given directory structure for your submitted articles:

```
A2
└── q2
    ├── forest_fire.sh
    └── report_q2.pdf
```

Time Limit

Each dataset will have a time limit of 60 minutes.

Your program must periodically write selected blocked routes to the output file so that partial credit may be awarded in case of timeout.

Directory Structure

```
A2
├── env.sh # For environment setup. This file may be empty.
├── q1
│   ├── Q1.py
│   ├── plot.png # generated from Q1.py
│   └── report_q1.pdf
└── q2
    ├── forest_fire.sh
    └── report_q2.pdf
```

For `env.sh` please only include `pip install` commands or c++ compilation commands (`make`, `gcc`, etc.). Please do not include any `conda` or `virtualenv` commands, we will take care of that.